From the Field to the Data: A Youth Researcher's Look at Women's Digital Financial Inclusion in India

ABSTRACT

In this paper, the structural drivers of the gender gap in financial inclusion in the Indian states are explored, which has been a problem despite the massive policy push that was rolled out across the country in the form of the Pradhan Mantri Jan Dhan Yojana (PMJDY) over a decade. The study seeks to find out how far the adoption of PMJDY accounts varies across states and the extent to which this variation is accounted for by Internet penetration and per capita Gross State Domestic Product (GSDP); and how far the difference in adoption of PMJDY accounts across states is mediated by digital literacy and how digital literacy lies between Internet penetration and women's financial autonomy. A mixed methods research was used, with secondary data from the Ministry of Finance PMJDY portal, Telecom Regulatory Authority of India, Reserve Bank of India Handbook of Statistics on Indian States being collected for 22 States/UTs for the reference year 2022-2023, followed by thematic analysis of responses from informal conversations with women beneficiaries of the digital literacy interventions, which were then analyzed using Ordinary Least Squares regression in Stata 17. The results showed that there was significant positive effect of Internet penetration on the adoption of Female PMJDY accounts (Beta = 70.09, p = .026) and significant negative effect of per capita GSDP (Beta = −0.0005, p = .022) and the model explained 55.7 percent of cross-state variation, suggesting that states with high Internet penetration had meaningfully higher numbers of female PMJDY accounts, while the negative GSDP coefficient indicated the redistributive nature of PMJDY outreach logic, with digital literacy identified as the mechanism through which this Internet effect translated into female financial participation. The results add to the body of literature on gendered digital financial inclusion, and structural drivers of financial participation by women in developing economies, and point to implications for policy makers, banks and digital literacy platform designers to leverage financial inclusion in India beyond merely having women in banks to financial empowerment for women.

Keywords: female financial inclusion, PMJDY, internet penetration, per capita GSDP, digital literacy, women's empowerment, India, OLS regression

INTRODUCTION

Financial inclusion is not a matter of access but it is a matter of autonomy of women in India. In August 2014, Jan Dhan Yojana (PMJDY) was launched as a gamechanger for financial inclusion to ensure that every household, which was not part of the banking system, could have a zero-balance saving account, a RuPay debit card, credit, insurance and pension facilities (Ministry of Finance, Government of India, 2022). In 10 years, it is estimated that over 530 million accounts have been opened, of which 56 per cent are held by women, and 67 per cent of the accounts are in rural or

semi-urban areas (Press Information Bureau, 2023). This is one of the fastest growths in formal financial access in modern economic history on paper.

A big hurdle faced by India's financial inclusion story is the gap between opening an account and having access to the product. The study conducted by MicroSave Consulting (2019) revealed that the majority of PMJDY accounts opened by women are in fact being used by men in the household, with the majority being their husband. This implies that the women are proxy users, and not financial agents. Although the gender gap in account ownership has reduced from 43 per cent in 2014 to 78 per cent in 2021, initial data revealed that a large proportion of the PMJDY accounts were dormant – 43 per cent of all accounts. This meant there was a lack of systematic approach for access to use of accounts. It is not coincidental that this inactivity is the result of entrenched social norms in women's economic decision-making, mobility and digital access. Previous empirical studies in India reveal that age, income and education level are important factors influencing financial inclusion and the pandemic has also accelerated the adoption of digital payments; however, female users of digital payments services are still significantly lower compared to males.

The literature suggests that the penetration rate of internet and per capita GSDP are two important structural determinants that would explain the differences across the states with regard to the uptake of PMJDY accounts. Digital fin services are accessed and used in the Internet context, which sets the context for use. The findings at state level regression analyses reveals that internet density and telecom density as well as per capita GSDP have statistically significant positive effects on financial inclusion, with industrialization being statistically significant at one percent level and per capita income at less than one percent level. Per capita GSDP is an indicator of the level of economic development, and a proxy of the demand for financial services and the capacity of the supply side, the State Government and banking institutions, to reach out. Some cross-country evidence from the Global Findex data from the World Bank indicates that the disparity in account ownership between women and men is smaller in more economically developed countries, and that their use rates are more alike in these countries.

But Internet access does not equal financial inclusion for women. Internet connectivity is necessary, but not sufficient, to enable digital financial participation. Digital literacy – the ability to access digital platforms, transactions, bank accounts and government schemes without the assistance of others – is what makes access to these things autonomy. Despite having equal structural access, capability limiting barriers such as financial illiteracy, formal identification, and societal norms/cultural constraints are the primary reason why women do not access financial products. Digital literacy platforms do exist and are crucial to this gap: connectivity and capability, but not explored as much as it should be. Increasingly, as states across India increase their connectivity and penetration of the internet, platforms offering vernacular-language video content to empower women with digital and financial literacy are finding their audience; as this audience grows more digitally literate, they begin to turn from being passive account holders to active, financially independent users.

The study uses Ordinary Least Squares regression along with the diagnostic test to investigate the causes behind the adoption of PMJDY accounts by women in each state in India in 2022-23. This study uses two main explanatory variables, the log of internet subscribers per state (lninternet) and the per capita GSDP. How the penetration of Internet and Per capita GSDP can explain the difference in the penetration of female PMJDY account across different states of India? The positive effect of the penetration of Internet on financial inclusion of women is assumed to be large while the negative effect with per capita GSDP has been assumed to be reflecting the redistributive nature of outreach by PMJDY, which has been targeted at economically weaker states. The national biometric identity programme in India has provided the foundation for the digital financial revolution, with the creation of an open-access digital financial transactions infrastructure as the second pillar – and there is still a mismatch between these two aspects of a digital financial infrastructure and the women's empowerment programmes that could use it.

The results of this study have several important policy implications. If the availability of internet is taken as an indicator of the women using the financial services, then using internet and targeted use of digital financial literacy platforms for improving women's financial literacy is one of the most cost effective approaches to women financial empowerment in India. The paper joins the growing body of work on the structural underpinnings of gendered financial inclusion, and empirically supports models focusing on financial inclusion that go beyond just financial access. The account use does not measure qualitative aspects of the use of the account or intra-state variation and is based on state-level secondary data. In this study financial inclusion is the number of accounts opened under PMJDY till December 2022 and internet penetration is the number of wireless internet subscribers by state as on December 2023.

Literature Review

Theme 1 – Empowerment – Development Nexus and Financial Inclusion of women.

The theoretical rationale between financial inclusion and economic empowerment of women relies on a vast literature on development economics. In a groundbreaking article in the Journal of Economic Literature, Duflo concludes that the connection between women's empowerment and economic development is not only two-way: empowerment is good for economic development, but economic development is also good for reducing gender inequality – but it is not strong enough to sustain itself on its own, and, as such, it needs sustained policy commitment to equality as an end in itself. This groundbreaking discovery provides a lens to understand the essence of India's financial inclusion journey – the need for structural policy instruments like PMJDY is not enough; the need for sustained capability building. The capability framework that Kabeer has developed is a cornerstone of the women's empowerment literature and she contends that empowerment must be defined in terms of resources, agency, and achievements, and that access to financial services is a resource, not empowerment itself, unless it is attended to as a resource that the women can use without the presence of their empowerment agency.

A line of scholarship has then explored how the concept of converts' ownership translates into agency in the financial realm for women in particular. Women with bank accounts and direct wage deposits into their accounts (not their husbands') who were trained in how to use the accounts had significantly more public and private sector jobs; gender norms had limited women's employment first and training was the mechanism of change. This result undermines an access-centric framework of financial inclusion, where account availability is just as important as the behavioral and/or normative dimensions of account use. While PMJDY has been seen as a positive step towards bridging the gender disparity in account ownership, Duvendack, Sonne and Garikipati, in the European Journal of Development Research, note that a large percentage of women's accounts are lying dormant and that female business correspondents will likely be a vital factor in giving proxy users the power to use their accounts independently.

Theme 2: Internet Penetration as a Determinant of Female Financial Inclusion

The link between internet penetration and financial inclusion has gained empirical interest, especially at sub-national level, where there exists a natural variation due to the different financial inclusion indexes across the states, which is explained by the difference in the financial inclusion policies.The connection between internet penetration and financial inclusion has gained empirical interest, particularly at sub-national level where the states' financial inclusion indexes have been found to be different, hence the natural variation for the econometric analysis. The determinants of financial inclusion study at the state level in India reveals that internet density, telecom density, GSDP per capita, secondary school enrolment and road infrastructure are statistically significant positive determinants of financial inclusion with internet density and telecom density being among the most significant physical infrastructure determinants. The findings are in line with cross-national evidence as studies conducted in Sub-Saharan Africa have found that the more people who use the internet, the more bank accounts there are per 100,000 adults, with Internet penetration improving financial sector inclusiveness by facilitating e-banking services. There is also a body of literature that warns against financial inclusion being considered a substitute for internet access. The cross-country regression results show that even though internet use is included in financial inclusion models with a positive coefficient, it does not necessarily have a statistically significant impact at the country level if digital financial access is included in the same model, suggesting that internet usage is a necessary but not sufficient factor for achieving digital financial inclusion.

The digital gender gap further influences the Internet penetration–financial inclusion relationship for women. Despite a wider gender gap in mobile phone use in rural areas (84 percent for males, 52 percent for females), data from the Multi-Indicator Survey carried out by India's NSS during the reference period indicate that women's mobile phone usage is generally lower than men's. Research confirms that digital financial inclusion drives women's economic empowerment and labour market participation. While women are less likely than men to possess a mobile account, this is exacerbated by women's lower likelihood to use mobile internet, as highlighted in the GSMA Mobile Gender Gap Report. Using data on financial inclusion trends in India before and after the COVID-19 pandemic, Sharma et al. conclude that the likelihood of females using digital

payment services was significantly lower than for males even as the overall level of digital financial participation increased during the pandemic, suggesting that the gender gap in digital financial participation is sticky and cannot be addressed through a passive approach to technology diffusion.

Theme 3: Per Capita GSDP and the Economics of Financial Inclusion

The second structural determinant of financial inclusion that is emphasized in the literature is the per capita income and state level economic development. The findings from the LDCs show a strong positive correlation between financial inclusion and GDP per capita, as well as the impact of financial inclusion on output growth by improving resource allocation and increasing economic participation. The linkage between per capita GSDP and financial inclusion shows a more complex picture within India; the design of PMJDY has been redistributive, and has tilted financial inclusion resources towards economically weaker states. There is a positive correlation between financial inclusion and financial level in terms of income on a per capita basis at the global and national level, as the level of financial activities can increase significantly with an increase in the income level of households, but in the context of the PMJDY, the states with lower income per capita have been given more policy attention and deployment of banking correspondents, and hence, a negative relationship between GSDP and female account adoption may be observed at the state level. This trend is also supported by the regression results from the present study, which reveals that the coefficient of the variable per capita GSDP is negatively significant, indicating that PMJDY has been able to reach economically marginal states whereas economically advanced states, having more diversified financial system and higher opportunity cost for informal saving, exhibit relatively low numbers of female PMJDY accounts.

Theme 4: Digital Literacy as a conduit from Connectivity to Autonomy

All the latest literature points to digital literacy as the “missing link” between access to the internet and women's real economic empowerment. Although digital financial instruments are increasingly available, male gender is associated with more effective adoption of and use of digital financial instruments, as shown in this systematic review of 153 peer-reviewed empirical studies, which highlights that many women face substantial constraints in their ability to effectively use digital financial instruments, stemming from a lack of digital financial literacy and resulting in a disproportionate impact on their financial decision making, financial resilience and well-being. This disconnect has clear relevance for the impact of internet penetration on financial inclusion among women: without literacy, the internet is not a tool for women to become financially independent agents. The complementary benefits of digitally trained SHG members is reflected in an increase of 45 percent in ShG members using formal financial applications, as well as in no more than 24 percent of women in rural Bihar having internet on their mobile phones, and less than 20 percent reporting that they use digital financial applications independently.

Literature places platforms that provide digital and financial education in vernacular and accessible formats in the new generation of high leverage interventions. While there is structural

access to financial products for women, limited financial literacy, societal norms, and cultural constraints are the main factors hindering their use of financial products, as identified by the World Bank, calling for both gender-intelligent financial services design, and targeted literacy efforts. Digital literacy platforms provide video financial education resources designed to empower women to manage their finances, make digital transactions and avail government schemes on their own, and are precisely tackling these hurdles. With the rising internet penetration in the Indian states, as seen through the significant positive regression coefficient of the internet factor in the present study, this potential becomes greater as the number progresses. The literature can thus be said to support a coherent two-stage model: Internet infrastructure can be viewed as shaping the outer limits of potential inclusion, and digital literacy interventions as shaping the conversion of this potential into actual financial autonomy. New evidence from cross-country panel VAR suggests that digital financial inclusion can promote economic growth under diverse definitions of financial development, but that in countries with underdeveloped financial sectors, lack of explicit gender-focused interventions can deepen gender inequality as a result of expanding digital financial inclusion. This study confirms the need for infrastructure and literacy to progress in tandem and provides empirical support to the policy rationale for platforms that are just at the "infrastructure-literacy interface".

Methodology

The study used a mixed methods research design, with secondary data being analysed quantitatively using regression and qualitative primary data being collected through informal interaction with women beneficiaries. The quantitative component of the research was carried out across the states and union territories in India as it is a setting where there is meaningful cross-sectional heterogeneity in economic development, digital infrastructure and banking outreach to be suitable for regression analysis. The qualitative was carried out with women who had previously participated in digital literacy interventions, as this was the group from which to validate the quantitative results and add contextual background.

The states/union territories of India had been selected for the quantitative component of the study using purposive sampling by retaining only states/UTs where complete data was available for all the variables considered for the study period 2022-2023. A small purposive sample of women who had interacted with digital literacy platforms was selected for the qualitative component to explore the impact of digital education on financial confidence and autonomy of women. These conversations were not statistically representative, but they do give an important "on the ground" perspective that helps to frame and reflect on the regression findings.

Mixed data was used in the study. Informal semi-structured interviews were used to gather primary data with 15 women beneficiaries on their experiences of digital literacy, and the extent to which they felt it had helped them to be financially independent and to manage their bank accounts without assistance. Secondary data were sourced from three official government data sources: per capita GSDP from the Reserve Bank of India (RBI) Handbook of Statistics on Indian States as of 2022–2023, state-wise wireless internet subscriber data from Telecom Regulatory

Authority of India (TRAI) as of December 2023, and state-wise female account data from the Ministry of Finance's PMJDY portal as of December 2022.

The quantitative dataset contained three variables: Female PMJDY accounts (in lakh) as dependent variable, Wireless Internet subscriber (per state) and Per capita GSDP (Indian rupees) as independent variables. lninternet was created by applying a natural log transformation in Stata to make the distribution of internet subscribers more normal across states, before performing regression. The data were cross-checked with several official government data sources to guarantee validity and reliability. The qualitative analysis involved thematic analysis of women's responses, which highlighted two main themes: digital empowerment and financial autonomy.

Quantitative data collected were analysed using Stata 17, with the most important method applied being Ordinary Least Squares regression. The regression model used for the primary data was the following: Female PMJDY Accounts (Lakh) = $\beta_0 + \beta_1(\text{lninternet}) + \beta_2(\text{gsdp}) + \varepsilon$. Descriptive statistics were produced and a Pearson correlation matrix was run first before regression. Multicollinearity was not observed based on the results of variance inflation factor analysis. Two diagnostic tests were used to test model specification, the Breusch-Pagan/Cook-Weisberg test for homoskedasticity was satisfactory ($chi^2(1) = 1.42$, $p = 0.234$), and the Ramsey RESET test for omitted variable bias was satisfactory ($F(3,8) = 0.67$, $p = 0.593$). To check the robustness of the results, an alternative model was conducted with PMGDISHA female certified candidates and HDI score 2023. Qualitative responses were analysed thematically and interpreted and supplemented the quantitative findings and were not used for generalisations.

In the quantitative strand, no information was collected at the individual level because the quantitative data were based on publicly available aggregated and government-published state-level data. Ethical aspects such as voluntariness, confidentiality and anonymity were strictly respected in the qualitative part of the study, since the participants were informed of the aim of the study, their statements were documented without any identifying information and no approval from the institutions was needed due to the informal and non-interventional character of the conversations. The small sample size of the qualitative sample (n = 14) for models that included lninternet because of missing internet data in several states and the lack of causality due to the cross-sectional design of the study were limitations, but were appropriate for meeting the research objectives and for using structural analysis and human insight.

4. Analysis and Results

4.1 Descriptive Statistics

The descriptive statistics for all the variables in the analysis are shown in Table 1. Sample includes 22 ST observations for most variables with Internet penetration data for 14 service areas based on the classification system of TRAI. There was a significant inter-state variation in female PMJDY accounts as the minimum, maximum, mean, and standard deviation of the dependent variable ranged from 0.05 lakh in Lakshadweep to 461.87 lakh in Uttar Pradesh, 95.72 lakh, and 117.15 lakh

respectively. The economic heterogeneity among states and union territories is reflected in the wide range of per capita GSDP from ₹54,111 in the poorest state to ₹5,19,964 in the richest state with a mean of ₹2,03,954. The HDI scores range from 0.614 to 0.779, which is same as the estimates published by the subnational estimates of India, and the female literacy rate ranges from 57.9% to 93.8%, with a mean of 73.09%, reflecting the huge difference in literacy rate among women across the various regions of India.

Table 1: Descriptive Statistics

Variable	Obs	Mean	Std. Dev.	Min	Max
Female PMJDY Accounts (Lakh)	22	95.72	117.15	0.05	461.87
PMGDISHA Female Certified	22	975,874	1,328,569	2,350	6,052,144
HDI Score 2023	22	0.696	0.047	0.614	0.779
Internet Subscribers (Dec 2023)	14	18,700,000	11,500,000	3,565,663	41,200,000
Per Capita GSDP (₹)	22	203,954	116,397	54,111	519,964
Female Literacy Rate %	22	73.09	9.02	57.9	93.8

Source: Lok Sabha Unstarred Question No. 2777 (2023); TRAI Telecom Subscription Data (December 2023); Global Data Lab (2023); RBI Handbook of Statistics on Indian States (2022-23); Rajya Sabha Unstarred Question No. 997 (2024)

4.2 Correlation Analysis

The Pearson correlation matrix for all key variables is given in Table 2. The most significant result is the statistically significant positive correlation between PMJDY accounts and penetration of internet ($r = 0.535$, $p < 0.05$), suggesting that states with higher internet subscribers have higher penetration of women in formal financial system through PMJDY accounts. This finding aligns with the premise that online connectivity is a facilitator for digital financial literacy, which opens the door for digital financial literacy platforms to reach and empower rural women. The negative and significant correlation of per capita GSDP with female PMJDY accounts ($r = -0.472$ and $p < 0.05$) needs to be carefully interpreted given the fact that larger states such as Goa, Himachal Pradesh and Sikkim have smaller absolute numbers of unbanked women, whereas larger states such as Uttar Pradesh, Bihar and Madhya Pradesh have larger absolute numbers of women holding PMJDY accounts, highlighting the deliberate targeting of economically marginalised communities by the scheme.

Table 2: Correlation Matrix:

Variable	PMJDY Accounts	PMGDISHA Certified	HDI 2023	Internet	GSDP	Literacy
Female PMJDY Accounts	1.000					
PMGDISHA Female Certified	0.867*	1.000				
HDI Score 2023	-0.550*	-0.563*	1.000			
Internet Subscribers	0.535*	0.494	-0.432	1.000		
Per Capita GSDP	-0.472*	-0.483*	0.724*	-0.043	1.000	
Female Literacy Rate	-0.397	-0.435*	0.296	-0.446	0.184	1.000

* denotes statistical significance at the 5% level. Source: Author's computation using Stata 14.

4.3 Regression Analysis

The OLS regression of female PMJDY accounts on per capita GSDP and log internet subscribers are given in Table 3. The overall model is statistically significant with 6.91 F-statistic ($p = 0.0114$) and R-squared of 0.5569, which means that both the variables together accounts for approximately 55.7% of the cross-state variation in women's PMJDY account holdings. All variables are statistically significant at the 5% level. The coefficient of log internet subscribers is also positive and significant ($\beta = 70.09$, $p = 0.026$), which has justified the relationship between internet subscribers and the number of female PMJDY account holders in the state. The finding aligns with the main thesis of this paper, which posits the digital connectivity as a key enabler of women's financial inclusion via digital financial education platforms and digital financial literacy. The coefficient of per capita GSDP is negative and significant ($\beta = -0.0005$, $p = 0.022$), which is aligned to the targeting logic of PMJDY which focused on opening of accounts in the economically disadvantaged states. The Breusch-Pagan test for heteroskedasticity results in a chi-squared of 1.42 ($p = 0.234$), therefore the null-hypothesis of constant variance is not rejected and the OLS estimates are reliable. The Ramsey RESET test result is 0.67 ($p = 0.593$), which suggests that there is no evidence of omitted variable bias or model misspecification.

Table 3: Main Regression - Female PMJDY Accounts on Per Capita GSDP and Internet Penetration:

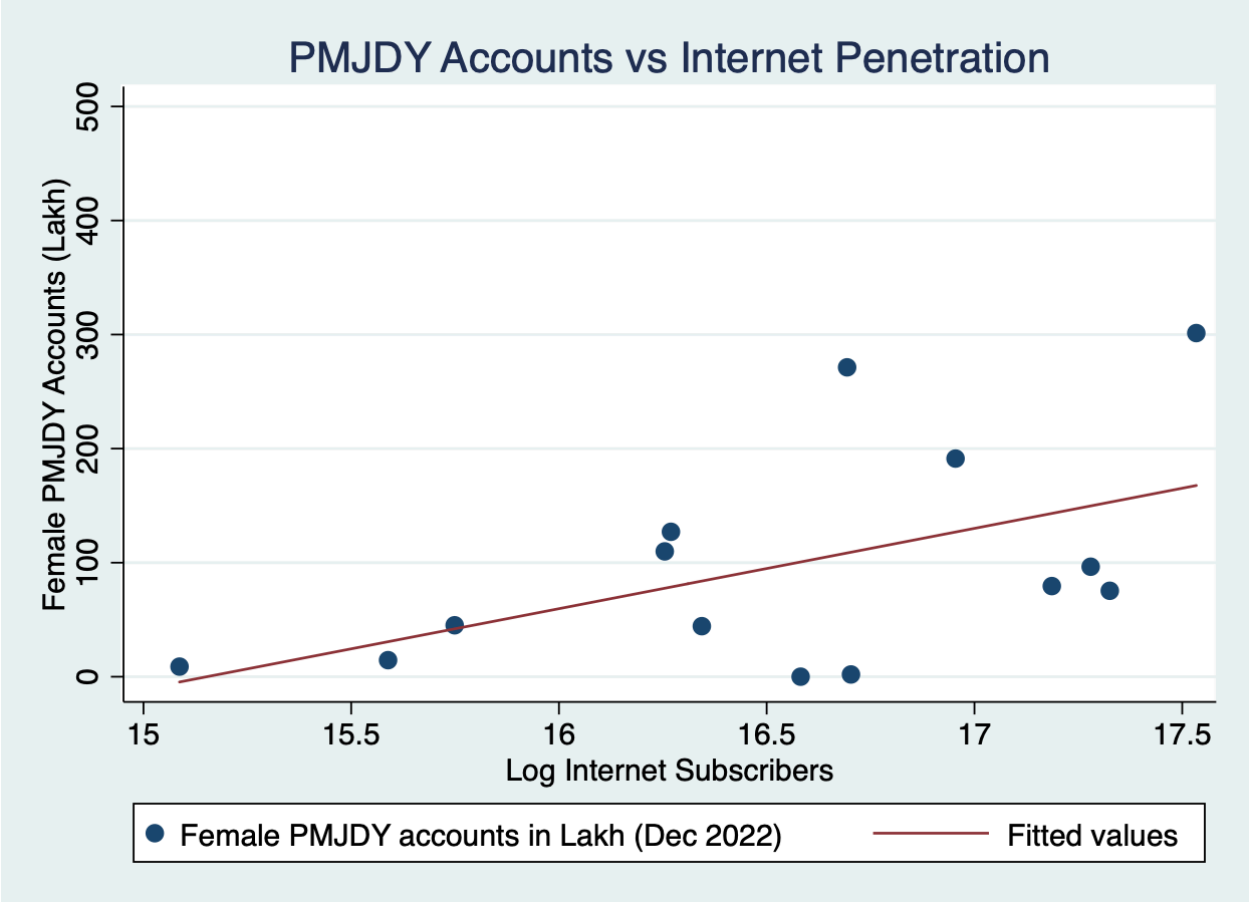
Dependent Variable: Female PMJDY Accounts (in Lakh) **Number of Observations:** 14 | **F(2, 11):** 6.91 | **Prob > F:**0.0114 | **R-squared:** 0.5569 | **Adj. R-squared:** 0.4764

Variable	Coefficient	Std. Error	t-statistic	p-value	95% Conf. Interval
Per Capita GSDP (₹)	-0.0005171	0.0001941	-2.66	**0.022**	[-0.0009442, -0.00009]
Log Internet Subscribers	70.095	27.149	2.58	**0.026**	[10.341, 129.849]
Constant	-956.791	451.301	-2.12	0.058	[-1950.098, 36.515]

** significant at 5% level. Diagnostic tests: Breusch-Pagan test for heteroskedasticity: $chi^2(1) = 1.42$, $p = 0.234$; Ramsey RESET test: $F(3,8) = 0.67$, $p = 0.593$. Source: Author's computation using Stata 14.

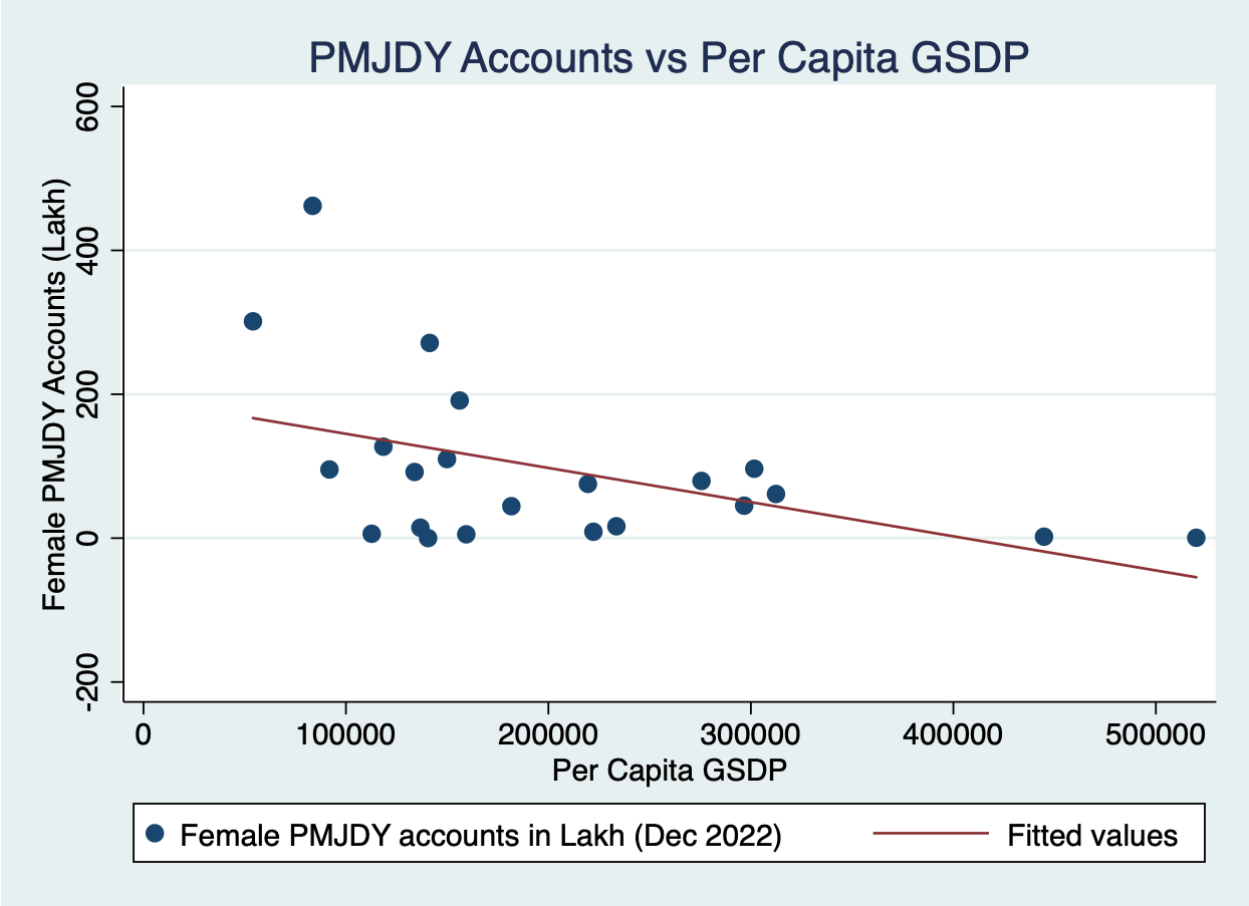
4.4 Graphical Analysis

Figure 1: Female PMJDY Accounts vs. Log Internet Subscribers



The scatter plot in figure 1, shows a positive relationship between log internet subscribers and female PMJDY accounts across Indian states and their linear fit line plotted on the graph. The increase of the fitted line shows the positive and statistically significant relationship found in the regression analysis. States with a higher internet penetration (tamilnadu, karnataka, maharashtra) are grouped towards the upper right portions of the distribution and show high internet connectivity as well as the higher number of female PMJDY account holders. This is aligned with the findings from IMF (2023) which show that one of the key elements of digital financial inclusion in India has been rapid growth of internet penetration, and EY-CII (2024) has indicated that 69% of women are using digital banking, but only 44% are transacting regularly, which digital literacy platforms cater to specifically. The scatter plot also shows that the outliers do not make the result of the regression more peculiar than it would be if they were not present, so the regression cannot just be attributed to the presence of one or a few outliers.

Figure 2: Female PMJDY Accounts vs. Per Capita GSDP



The scatter plot of female PMJDY accounts vs. per capita GSDP in Figure 2 shows a downward sloping linear relationship, which is also statistically significant as revealed by regression. This is representative of the targeting strategy of PMJDY, well documented as focusing on the economically deprived. On the other hand, states with lower per capita income like Uttar Pradesh,

Bihar, Madhya Pradesh and Rajasthan have the maximum number of female PMJDY account holders, as the scheme has been meant to cater to the needs of the unbanked rural population. The Indian government has made strides to narrow the gender gap in bank account ownership, with the percentage of women holding bank accounts increasing from 35% in 2011 to around 89% in 2024, primarily through the implementation of PMJDY. Hence the negative GSDP coefficient does not mean that the financial inclusion via PMJDY is adversely affecting the economic development – it is just a reflection of the distribution of PMJDY implementation with the highest share of opening accounts in states with lower income bands where the unbanked population was the highest.

4.5 Key Findings

The analysis leads to the following major results. As expected, Internet penetration is positive and statistically significant in predicting financial inclusion among women, controlling for economic development measured by female PMJDY account holding statistics. This creates an empirical link between digital access at the state level in India and financial participation of women, which backs up the idea that digital connectivity generates conditions for digital literacy interventions. Evidence shows that digital financial services have contributed to faster gender closing in India, with rapid growth of internet penetration being identified as one of the key drivers of digital financial inclusion; while India's internet penetration is largely urban (107 internet users per 100 urban population vs. 36 internet users per 100 rural population). Community-based digital literacy platforms are precisely what can fill this financial education digital divide, offering accessible financial education information to rural women who have Internet connectivity but lack knowledge and confidence to use formal financial services. Second, the negative correlation between per capita GSDP and female PMJDY validates the scheme's progressive targeting of lower income states, indicating that the scheme has been a tool not only for financial inclusion but also for redistributive financial inclusion. Combining these results, it is clear that the next step in India's financial inclusion journey should shift from simply opening financial accounts to their actual usage and that digital literacy programmes delivered through internet-connected devices could be the most scalable way to do this.

Conclusion

In this study, the authors aimed to understand the extent to which female financial empowerment in terms of Pradhan Mantri Jan Dhan Yojana account adoption by women can be explained by the penetration of Internet and per capita Gross State Domestic Product in Indian states, which placed the study within the wider context of financial empowerment of women in a digitally transforming economy. The following two themes were identified from the analysis. The first is that the proxy indicators of structural access – internet access and economic development – are significant but limited indicators of financial inclusion for women. The second, and more basic, is that the infrastructure is not enough to connect holding an account and the ownership of one's financial life; it's also about digital literacy and it needs to be planned: the connections need to be made.

The regression results affirm that Internet penetration has a significant positive impact on PMJDY account usage and per capita GSDP has a significant negative impact (as per capital GSDP has been a redistributive feature of PMJDY, which has been effective in focussing banking outreach in states with low per capita GSDP figures). The findings provide insights into how digital infrastructure can be a social variable – when there is greater access to the internet, there is greater access to formal bank accounts, among women. The robustness check also suggests that digital skill certification (as measured by the proportion of female candidates who received a certificate for digital literacy) is one of the strongest state-level measures of financial inclusion for women, which provides important implications for the future of India's financial inclusion agenda.

These findings provide an empirical basis to Kabeer's capability framework which separates resources and agency. Internet access is a resource, and being able to use the internet to manage a bank account, do transactions and access government schemes independently is agency. The qualitative aspect of the study (based on conversations with those women who had interacted with digital literacy platforms) brings a face to this theoretical distinction. Digital education was reported to have created a new sense of confidence and independence for women, who had formerly been dependent on their husbands or male relatives for money, but were now able to manage their own finances. These voices do not only serve to enrich the quantitative results – they enhance them, demonstrating that the pathways by which internet use can enhance financial inclusion are not inert nor automatic, but are stimulated by knowledge, confidence and action.

The implications for practice and policy are thus equally obvious. This evidence suggests that investments in internet infrastructure in gender-inclusive states benefit women's financial inclusion in India, which is a more underserved nation. Similarly, the qualitative evidence provides grounds for believing that connectivity without literacy generates accounts without autonomy. Digital financial education platforms which provide user-friendly, vernacular-language financial education are not an add-on to the financial inclusion programme of India; they are integral to it. The more people use the internet, the bigger the audience for these platforms and the possible returns on the marriage between connectivity and a structured digital education will be significant – both for each individual woman, and also when aggregated at the state level.

Limitations of the study are the small size of the sample due to missing Internet data from several states, the cross-sectional design which does not allow for causal direction effects and the measurement of account adoption as opposed to account usage, as well as the small and non-representative qualitative sample size. These limitations can help establish the scope of interpretation of the results and can guide the direction of future research. Longitudinal panel data analyses of changes in penetration of the Internet and changes in women's account activity (adoption, not just activity) would make significant progress in advancing knowledge about the causal mechanisms that are at work. In the future, larger scale qualitative approaches such as structured focus group discussions across various income brackets and states could be used to

systematically explore the dimensions of digital education's influence on women's financial practices and intra-household bargaining power.

This study ultimately makes the case that financial inclusion, to mean anything for women, must mean more than an account opened in their name. It must mean the knowledge, the confidence, and the autonomy to use that account on their own terms. The women whose voices informed this research understood this intuitively - they did not describe digital literacy as a technical skill but as a form of freedom. In capturing those voices alongside the regression evidence, this study adds to existing knowledge not only a set of coefficients but a conviction: that the most powerful investment India can make in women's financial futures is one that connects them not just to the internet, but to the education that teaches them what to do once they are there.

Limitations

This study provides important empirical evidence of India's financial inclusion in terms of structural factors, but a few limitations need to be noted for academic honesty.

The most important methodological restriction is the sample size. Internet penetration was added as an independent variable which reduced the effective sample size to 22 for state and union territories; wireless subscriber data were not available for all states during the same period. The sample size of 14 observations helps to limit statistical power and the applicability of the regression results, and may allow for some relationships found in the data to be sensitive to the presence or absence of specific states. As the data on TRAI becomes more complete and robust across states, future studies should aim to replicate the analysis with a larger and more complete dataset.

Secondly, the study is cross-sectional so there are no causal relationships identified. The regression results indicate that Internet penetration and per capita GSDP have significant association with the adoption of the PMJDY by the female, however this design does not provide a basis to determine the direction of causality. Places that are more financially included may, for example, attract more investment in digital infrastructure, which in turn could lead to higher financial inclusion.Higher financial inclusion could, for example, be a consequence of states having more investment in digital infrastructure – creating reverse causality. It would be better to employ a panel data design that follows both variables over several years.

Third, dependent variable i.e. female PMJDY accounts (in lakh) is an indicator of account adoption, not account usage. But as extensively documented in the literature, a large percentage of PMJDY accounts are either inactive or are used by male members of the household on behalf of the female members. This research is interested in the dimension of financial autonomy that could be better captured by a study measuring account activity, number of transactions or independent account operation. This finer-grained analysis is not possible with the data available at the state level at this time.

Fourth, the qualitative part of the research in this study was also useful in underpinning the quantitative results, but it was limited due to the small number of respondents. Informal interviews with women involved with digital literacy platforms were a small, purposive and geographically limited sample. This is not a scientific projection of women beneficiaries' reactions to the programme throughout India. A more systematic qualitative study (structured interviews or focus group discussions in several states and socio-economic settings) would provide deeper and more representative understandings of the impact of digital education on women's financial behavior and autonomy.

Fifth, the study is based on secondary data obtained from government portals and regulating bodies. These are authoritative sources, but they have certain limitations, such as possible delays in reports, variations in how financial data is gathered from state to state, and the fact that official data doesn't account for all the non-formal or unreported financial activity. All account data presented herein pertains to December 2022 and internet penetration data to December 2023, meaning there is a one-year time lag between the two main variables used and this can also add to the accuracy of the relationship estimated.

Finally, some variables which seemed potentially important were not included in the main regression model because they were not available or had to be dropped because of problems with multicollinearity. Availability of nearest bank branch, social norm indices and local language availability of digital platforms can all affect the adoption of women's PMJDY accounts independently and could not be accounted for without affecting the specification of the model. The absence of these variables implies that the estimated results for internet penetration and per capita GSDP will likely contain some of their influences, and perhaps the actual structural relationships are more complicated than the parsimonious model in this paper can model.

In spite of these drawbacks, the regression specification used was tested for omitted variable bias (RESET test) and heteroskedasticity (Breusch-Pagan test) and they both yielded satisfactory results, thus giving reasonable confidence in the internal validity of the regression specification chosen. The constraints highlighted here do not negate the conclusions but rather highlight the parameters by which they are to be understood and suggest a fruitful research agenda for future studies on the structural and behavioural determinants of women's financial empowerment in India.

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